BrightCart

BRIGHTCART

Architecture

VERSION	1.0.0
DATE	2026-10-18
AUTHOR	BrightCart team

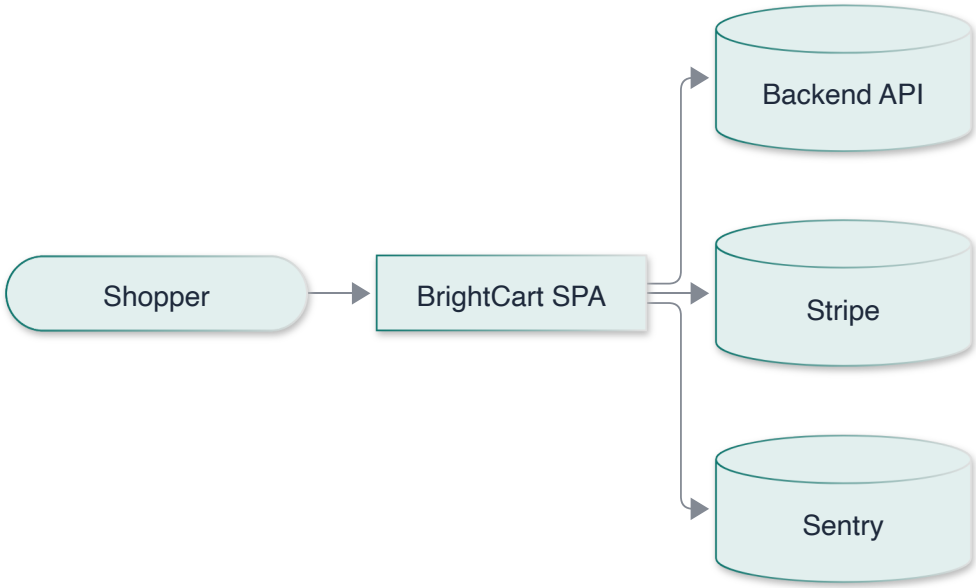
Contents

Overview 3
System context 3
Components 3
Data flow: checkout 4
Deployment view 4
Cross-cutting concerns 4
Decision log 4
ADR-001: Single-page app on Vite 4
ADR-002: Tailwind CSS for styling 5
ADR-003: Vercel hosting 5
Related documents 5
Assumptions and open questions 5

Overview

BrightCart is a client-side React single-page application. Vite builds it into static files that Vercel serves. The browser talks directly to three external systems: BrightCart's own backend API (`VITE_API_URL`), Stripe for payments and Sentry for error reporting. There is no server code in this repository.

System context

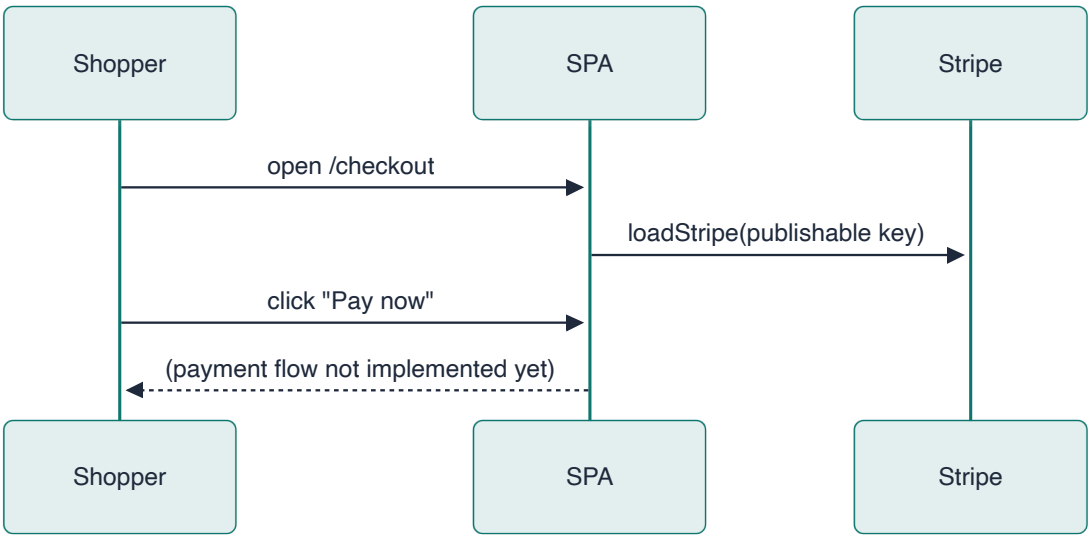


External system	Used for	Evidence
Backend API	Product data (planned)	`VITE_API_URL` read in `src/pages/Home.tsx`
Stripe	Payments	`loadStripe` in `src/pages/Checkout.tsx`
Sentry	Error reporting	`Sentry.init` in `src/main.tsx`
Vercel	Hosting and deploys	`vercel.json`, `deploy` job in `.github/workflows/ci.yml`

Components

Component	Responsibility	Tech	Location
App shell	Router and page layout	React Router 6	`src/App.tsx`
Home page	Product listing	React, Tailwind	`src/pages/Home.tsx`
Checkout page	Starts payment	Stripe.js	`src/pages/Checkout.tsx`
ProductCard	Product tile	React	`src/components/ProductCard.tsx`
Bootstrap	Mounts React, starts Sentry	React DOM	`src/main.tsx`

Data flow: checkout



1. React Router renders `Checkout` for `/checkout` (`src/App.tsx`).
2. Importing the page calls `loadStripe` with `VITE_STRIPE_PUBLISHABLE_KEY`.
3. The **Pay now** button has no handler yet, so no payment is created ASSUMPTION that this is work in progress.

Deployment view

GitHub Actions runs `lint` and `test` on every push and pull request. On `main` it also runs `npx vercel deploy --prod` with the `VERCEL_TOKEN` secret. `vercel.json` rewrites every path to `/`, so client-side routes survive a page refresh.

Cross-cutting concerns

Concern	Approach	Evidence
Configuration	Build-time `VITE_*` env vars	`.env.example`
Error handling	Sentry browser SDK	`src/main.tsx`
Styling	Tailwind with brand tokens	`tailwind.config.js`
Authentication	None in the frontend	no auth code found

Decision log

ADR-001: Single-page app on Vite

- **Status:** Accepted

- **Context:** The storefront needs fast navigation and no server rendering ASSUMPTION.
- **Decision:** React 18 + Vite, routed client-side (`package.json`, `src/App.tsx`).
- **Consequences:** Static hosting is enough. SEO for product pages is limited without pre-rendering.

ADR-002: Tailwind CSS for styling

- **Status:** Accepted
- **Context:** One consistent brand palette across components.
- **Decision:** Tailwind 3 with brand colours and fonts in `tailwind.config.js`.
- **Consequences:** No separate CSS files per component; styles live in class names.

ADR-003: Vercel hosting

- **Status:** Accepted
- **Context:** Static SPA with preview needs ASSUMPTION.
- **Decision:** Deploy with the Vercel CLI from CI (`.github/workflows/ci.yml`).
- **Consequences:** Deploys depend on the `VERCEL_TOKEN` secret and Vercel availability.

Related documents

Developer Guide · User Guide · CI/CD and Deployment

Assumptions and open questions

#	Item	Type	Why unverified	Who can confirm
A1	Pay now is unfinished work, not a bug	assumption	button has no handler	frontend lead
A2	Reasons behind ADR-001 and ADR-003	assumption	inferred from code, no ADR files	tech lead
Q1	What API does `VITE_API_URL` point to, and who owns it?	open question	backend not in repo	engineering lead